WEEK 8 · LECTURE L2 · UNIT III

Storytelling with Data, and the Ethics of Framing

How a chart becomes a decision, and how a true chart, framed one way, can still mislead.

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

1. Structure a data argument: three acts (what, so what, now what) and the big idea, evidence, caveat, ask.
2. Apply Knaflic’s six lessons and match the chart to the message.
3. Explain System 1 and System 2 (Kahneman) and framing effects (Tversky and Kahneman), and why a true chart can mislead.
4. Apply Heath and Heath, Huff and Pandey et al., and tailor a story to each audience ethically.

ONE SOURCE

The same theories as the in class lecture and the course hub, without the simulation data used in class. Course outcomes: CLO 4 and CLO 5.

COMMIT BEFORE THEORY

Commit to an answer first

Two slides report the same result: “Training cut attrition from 20% to 15%” and “Attrition fell by 25% after training”. What is true?

A

The first is honest and the second is false

B

The second is wrong because 20 minus 15 is 5

C

Both are true; one reports percentage points and one percent change, and each can steer the reader

D

Both mislead because attrition cannot be measured

EXAM LENS

The answer is C: 5 points is a 25% relative fall.

KNAFLIC (2015)

Six lessons for storytelling with data

1 · Context

Who is the audience (narrow it to the decision maker), what must they know or do? Separate exploratory analysis (opening 100 oysters) from explanatory communication (showing the two pearls).

2 · The right visual

Simple text for one or two numbers; tables or heat maps for mixed audiences; scatter for relationships; lines and slopegraphs for time; horizontal bars for categories. Bars start at zero. Avoid pies and donuts.

3 · Remove clutter

Clutter adds cognitive load.

4 · Focus attention

Use Gestalt principles and preattentive attributes on the point.

5 · Think like a designer

Form follows function.

6 · Tell a story

Beginning, middle, end; repetition and flow make it stick.

EXAM LENS

Begin with the big idea (after Nancy Duarte): one complete sentence with a point of view and what is at stake; then a three minute story; storyboard on paper first. Knaflic built Google People Analytics' visualization course.

THE STRUCTURE OF A DATA ARGUMENT

Three acts, and four parts

Setup: what?
The question and the current state.

Conflict: so what?
The gap the data reveal.

Resolution: now what?
The evidence based action.

The big idea
What is true and why it matters, in one sentence.

The evidence
One or two charts that prove it, honestly framed.

The caveat
What the data cannot show.

The ask
The decision you want, with owner and date.

EXAM LENS

A data story without an ask is a report; one that hides its caveat is advocacy. Give every number a comparison: over time, against peers, target or scenario. Most analytics fails at recommendation and decision, the gap storytelling closes.

MATCH THE CHART TO THE MESSAGE

And the audience

Message	Chart
Comparison	Bar or column
Trend over time	Line
Relationship	Scatter
Distribution	Histogram or box plot
Composition	Stacked bars; pies only rarely

Audience	What they need
Board, executives	Impact, risk, ROI; one or two slides
HR leaders	Drill downs, root causes, leading indicators
Line managers	Simple team views, plain language, actions
Employees	Personal insight, privacy protected, no stigmatizing labels

Course’s Data Storytelling teaching note. The note allows pies “judiciously”; Knaflic advises avoiding them.

KAHNEMAN (2011)

Why stories work: two systems of thinking

System 1

Fast, automatic, intuitive; reads a clear visual at once; accepts a vivid chart before checking it.

System 2

Slow, effortful, deliberate; follows a narrative argument; checks the evidence, if prompted.

EXAM LENS

Show the point to System 1; give System 2 the reasons and the caveats. Numbers without a story demand System 2 effort busy executives rarely give.

TVERSKY AND KAHNEMAN (1981)

Framing changes choices

Frame	Chose the sure option	Chose the risky option
Lives saved (gain frame)	72%	28%
Lives lost (loss frame)	22%	78%

EXAM LENS

Same options, different frame: gain frames invite caution, loss frames invite risk taking. In HR: retention 85% or attrition 15%; points or percent change; gain or loss. Each frame steers the reader. Use framing to discuss the ethics of presentation, not only accuracy; experts are not immune.

Percentages as reported in the original study (the “Asian disease” problem).

A TRUE CHART CAN STILL MISLEAD

Honest visualization

Counts or rates

Operations has the most exits; Technology the highest rate. Both true: say which you show and why.

Truncated axes

A bar axis not starting at zero inflates small differences.

Cherry picked periods

Start and end dates that flatter the trend; show limits and contrary evidence.

Area and 3D distortion

Bubbles and 3D bars exaggerate size.

Missing base

Percent change without the starting level.

Hidden uncertainty

Small groups shown as precisely as large ones; show ranges and sample sizes.

EXAM LENS

Deceptive techniques measurably change how viewers read charts (Pandey et al., 2015). An honest chart lets the reader reach the right conclusion without the presenter in the room. Privacy and dignity: aggregates must not allow reidentification; avoid labels such as “flight risk” or “deadwood” for people.

THEORY IN DEPTH · HEATH AND HEATH (2007)

Ideas stick when they are simple and concrete

The core idea

SUCCESs: Simple, Unexpected, Concrete, Credible, Emotional, Stories.

How it works

A concrete case and an unexpected fact carry an abstract number into memory.

Limits and critique

Stickiness serves bad ideas too; it is a tool, not a test of truth.

IN PRACTICE

One concrete employee story can make an abstract completion gap memorable for a Board.

EXAM LENS

Pair stickiness with honesty: a memorable story must still be representative.

THEORY IN DEPTH · HUFF (1954)

How to lie with statistics

The core idea

A classic catalogue of tricks: biased samples, well chosen averages, missing figures, gee whiz graphs.

How it works

Five questions: who says so, how do they know, what is missing, did somebody change the subject, does it make sense?

Limits and critique

Old examples, but the five questions remain a sound checklist.

IN PRACTICE

Apply Huff’s five questions to every vendor claim about adoption or ROI.

EXAM LENS

Huff’s questions are a ready made structure for evaluating any statistic in an exam case.

APPLY

Two stories from one survey

Teaching case (Course framing)

An engagement survey at a Kolkata and Delhi media group shows favorable responses fell from 62% to 55%. One manager's slide says "Engagement collapsed" over a bar chart whose axis starts at 50%. Another says "More than half our people remain engaged". The Kolkata office has only 40 respondents.

Work it through

1. Compute the change in points and in percent.
2. Which techniques make the first slide misleading?
3. How does each slide use framing?
4. Write an honest big idea in one sentence, with a caveat.

CHECK YOUR ARITHMETIC

62% to 55% is a fall of 7 percentage points, or about 11.3 percent (7 ÷ 62).

COMMON CONFUSIONS

Tempting readings, and better ones

A tempting reading	A better reading
Percentage points and percent change are the same	From 20% to 15% is 5 points but a 25% relative fall; say which you mean
If every number is true, the chart is honest	Axes, periods and frames can mislead with true numbers
A good story leaves out the caveats	The caveat makes the ask credible
A sticky story is a true story	Stickiness is a tool; representativeness is the test

COMPARE AND EVALUATE

Six lenses on data storytelling

Lens	Use it to	Watch out for
Knaflic (2015)	Design the story and the visual	Practitioner experience, not experiments
Kahneman (2011)	Address both systems	Popular synthesis of many studies
Tversky and Kahneman (1981)	Choose and disclose a frame	Effect sizes vary
Heath and Heath (2007)	Make findings memorable	Stickiness serves bad ideas too
Huff (1954)	Interrogate a statistic	Dated examples
Pandey et al. (2015)	Show deception has effects	Laboratory setting

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse

An HR business partner must present a fall in first year attrition from 30% to 24% to a sceptical Board. Using Knaflic, Kahneman and Tversky and Kahneman on framing, design the presentation and explain how it avoids misleading the Board. **20 marks**

A strong answer: states the big idea and computes both 6 points and the 20% relative fall; chooses an honest visual and explains axis and comparison; explains System 1 and 2 and the frame; includes a caveat and an ask, and one remaining risk.

"A true chart can still mislead." Discuss with reference to framing effects and honest visualization principles. **20 marks**

A strong answer: uses Tversky and Kahneman; counts versus rates; scales and uncertainty; Huff's questions; proposes standards.

Apply Huff's five questions to a vendor claim that its app "raised engagement by 30%". **10 marks**

A strong answer: applies each question concretely.

CARRY THESE INTO THE EXAMINATION

Four things to remember

1. **Every data story needs what, so what and now what.**
   Big idea, evidence, caveat, ask.

2. **Say points or percent, counts or rates.**
   Each frame steers the reader.

3. **An honest chart shows its limits.**
   Zero baselines, full periods, uncertainty.

4. **Persuasion has an ethical limit.**
   Memorable must still be representative.

SET READINGS · FROM THE LESSON PLAN

Read for this lecture

- Knaflic, C. N. (2015). *Storytelling with data*. Wiley.
- Kavanagh and Johnson (2018), Chapter 14; Angrave et al. (2016).

NOTE

The other works named in this lecture are theory sources for reference, not set readings.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

- Heath, C., & Heath, D. (2007). *Made to stick*. Random House.
- Huff, D. (1954). *How to lie with statistics*. W. W. Norton.
- Kahneman, D. (2011). *Thinking, fast and slow*. Farrar, Straus and Giroux.
- Knaflic, C. N. (2015). *Storytelling with data: A data visualization guide for business professionals*. Wiley.
- Pandey, A. V., Rall, K., Satterthwaite, M. L., Nov, O., & Bertini, E. (2015). How deceptive are deceptive visualizations? *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, 1469 to 1478. https://doi.org/10.1145/2702123.2702608
- Tversky, A., & Kahneman, D. (1981). The framing of decisions and the psychology of choice. *Science, 211*(4481), 453 to 458. https://doi.org/10.1126/science.7455683